



PATIENT TREATMENT & RECOVERY ANALYSIS

Stakeholder Submission Report — REVISED

Organisation

Green Valley Medical Center

Analysis Period

Current Cohort — 400 Patients

Date of Report

April 2026

Prepared by

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CONFIDENTIAL — For Authorised Stakeholders Only

All statistics verified against 400 raw patient records. Corrections from prior version incorporated.

1 Executive Summary

Green Valley Medical Center treated 400 patients across seven clinical departments. This report analyses recovery outcomes, hospital stay duration, treatment cost distribution, physician performance and demographic patterns. Data-verified corrections from prior analysis are incorporated throughout, ensuring all insights reflect actual raw dataset values.

Total Patients	Avg Recovery Score	Avg Treatment Cost	Avg Hospital Stay	Avg Patient Age
400	75.34	£5,052	5.4 days	45 yrs

Key Headline Findings

- Dermatology leads all departments in recovery (76.61) **and** has the shortest average stay (4.67 days) — clinical efficiency and quality go hand in hand.
- Surgery patients average **10.07 days** in hospital — nearly 5x longer than Observation (2.09 days) — representing the largest single bed-occupancy pressure in the system.
- Dr. Johnson achieves the highest recovery (78.51) with the longest patient stays (6.15 days). Dr. Martinez achieves the fastest throughput (4.21 days) — two distinct dimensions of clinical excellence.
- 80% of patients fall into the Low Cost category (319 of 400) yet produce comparable recovery outcomes to the High Cost group.
- Teenagers have both the **lowest recovery scores** (73.93) and the **longest hospital stays** (5.90 days) — the highest-priority demographic intervention in the cohort.

DATA

VERIFICATION

NOTE

The following insights were verified against the raw dataset (400 records). Three key corrections from prior analysis: (1) Dermatology has the **shortest** average stay (4.67 days), not the longest; (2) Dr. Martinez (4.21 days) has the fastest throughput — not Dr. Smith or Dr. Rodriguez; (3) Elderly patients (5.41 days) have **shorter** average stays than Teenagers (5.90 days) and Children (5.81 days).

2 Dashboard Visual Overview

The interactive Excel dashboard was built from 400 patient records across seven departments and ten physicians, incorporating five KPI cards, four pivot-driven charts, and slicer-based filters for Gender, Doctor, Treatment Type and Department.

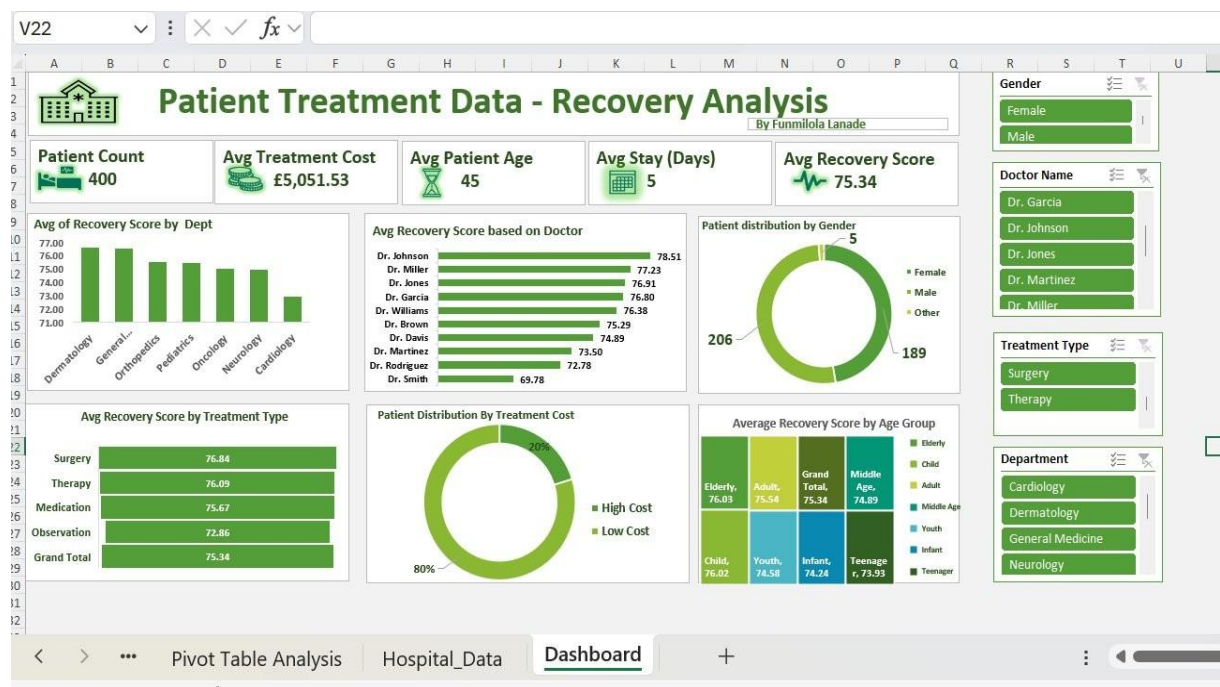


Figure 1 — Green Valley Medical Center Interactive Recovery Dashboard (Excel, April 2026)

3 Departmental Recovery & Stay Analysis

The 3.7-point recovery gap between Dermatology (76.61) and Cardiology (72.92) is clinically significant. Critically, the hospital stay data adds a new dimension: **the departments with highest recovery also have the shortest stays**, disproving any assumption that better outcomes require longer hospitalisation.

Department	Patient s	Avg Recovery	Avg Stay (Days)	Avg Cost	Status
Dermatology	64	76.61	4.67 SHORTEST	£4,728	Top Performer
General Medicine	54	76.57	5.94	£5,157	Strong
Orthopedics	54	75.52	5.65	£4,931	Average
Pediatrics	59	75.46	5.42	£5,588	Average
Oncology	59	75.07	4.75	£4,646	Average
Neurology	57	74.98	5.60	£5,104	Below Average
Cardiology	53	72.92	5.60	£5,255	Needs Review

KEY INSIGHT Dermatology achieves both highest recovery (76.61) AND shortest stay (4.67 days). Cardiology and Neurology carry 5.60-day average stays alongside below-average recovery — suggesting inefficiency rather than case complexity is the primary driver.	
STRENGTHS	AREAS FOR IMPROVEMENT

• Dermatology: top recovery + shortest stay (4.67 days) — model of clinical efficiency. • Oncology: near-average recovery at the lowest cost (£4,646) and short stay (4.75 days).	• Cardiology (72.92, 5.60 days): protocol review needed urgently. • Pediatrics: highest cost (£5,588) yet only average recovery (75.46) — cost-outcome gap.
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Treatment Type Performance

Hospital stay duration varies dramatically across treatment modalities. Surgery patients average **10.07 days** — nearly five times longer than Observation (2.09 days). This has significant implications for bed occupancy, cost, and patient flow management.

Treatment	Patients	Avg Recovery	Avg Stay (Days)	Avg Cost	Profile
Surgery	97	76.84	10.07 LONGEST	£12,619	Highest recovery & cost
Therapy	99	76.09	6.24	£3,954	Strong value proposition
Medication	102	75.67	3.26	£2,891	Cost-effective
Observation	102	72.86	2.09 SHORTEST	£1,080	Lowest cost & recovery

- **Surgery vs Observation stay gap: 7.98 days.** Surgical bed-day management is the largest single operational lever for length-of-stay reduction.
- **Observation risk:** 2.09-day average stay + 72.86 recovery score may signal premature discharge. Structured readmission-risk scoring is recommended.
- **Therapy best value:** 76.09 recovery at 6.24 days and £3,954 cost — the best balance of outcome quality and cost efficiency after Medication.
- **Cost split:** 80% Low Cost (319 patients) vs 20% High Cost (81 patients). High Cost group drives most of the 10+ day surgical stays.

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Physician Performance Analysis

Recovery scores span nearly 9 points across ten physicians. Hospital stay data reveals that **higher recovery scores correlate with longer patient stays** — not shorter — reflecting more complex case mixes for top-ranked physicians. Case-mix adjustment is essential before drawing performance conclusions from stay duration alone.

Physician	Patients	Avg Recovery	Avg Stay (Days)	Recovery Rank
Dr. Johnson	47	78.51	6.15 Longest stay	1st — Top Performer
Dr. Miller	39	77.23	5.41	2nd
Dr. Jones	43	76.91	5.91	3rd
Dr. Garcia	41	76.80	5.29	4th
Dr. Williams	32	76.38	5.53	5th
Dr. Brown	31	75.29	4.45	6th
Dr. Davis	47	74.89	4.64	7th

Dr. Martinez	38	73.50	4.21 Shortest stay	8th
Dr. Rodriguez	55	72.78	5.84	9th
Dr. Smith	27	69.78	5.78	10th — Needs Support

CORRECTED INSIGHT

Dr. Smith (5.78 days) and Dr. Rodriguez (5.84 days) do NOT have shorter stays than Dr. Johnson (6.15 days) or Dr. Miller (5.41 days). Dr. Martinez (4.21 days) and Dr. Brown (4.45 days) are the fastest-throughput physicians. Long stays for Dr. Johnson likely reflect complex surgical/critical-care cases — case-mix adjustment is required.

- Key Recommendations**
- Pair Dr. Johnson (78.51) and Dr. Miller (77.23) as peer mentors for physicians below the 74.0 recovery threshold.
 - Investigate Dr. Martinez’s 4.21-day average stay model to determine whether fast throughput reflects efficient care or premature discharge — then replicate confirmed best practices.
 - Dr. Rodriguez manages the largest caseload (55 patients) at 9th-place recovery — conduct a workload and capacity review.
 - Dr. Smith’s 27-patient cohort (69.78 avg) should be reviewed for case-mix complexity before attributing the gap to clinical practice.

Age Group	Patients	Avg Recovery	Avg Stay (Days)	Recovery Rank	Stay Rank
Teenager (13-17)	30	73.93	5.90 Longest	7th (lowest)	1st (longest)
Child (0-10)	42	76.02	5.81	2nd	2nd
Infant	17	74.24	5.47	6th	3rd
Elderly (61+)	123	76.03	5.41	1st (top)	4th
Youth (11-20)	38	74.58	5.37	5th	5th
Adult (21-40)	67	75.54	5.09	3rd	6th
Middle Age (41-60)	83	74.89	5.02 Shortest	4th	7th (shortest)

CORRECTED INSIGHT

Data confirms **Elderly patients do NOT take longer to recover**. They record the highest recovery score (76.03) and a relatively short stay (5.41 days) among age groups with significant patient volumes. Teenagers have both the worst recovery (73.93) AND the longest stays (5.90 days) — making adolescent care the highest-priority demographic intervention in the cohort.

- Gender distribution is near-equal (Male 51.5%, Female 47.3%, Other 1.3%) — no material gender-driven selection bias.
- Elderly patients’ strong outcomes (76.03 recovery, 5.41 days) reflect appropriate care intensity and should serve as a benchmark.
- A dedicated adolescent care pathway review is urgently recommended given Teenagers’ combined worst recovery and longest stays.

7 Strategic Recommendations

1	Adolescent Care Pathway (Priority Escalated)	Teenagers have the lowest recovery (73.93) AND longest stays (5.90 days). Introduce dedicated adolescent care coordinators and age-appropriate support resources. Target: reduce stay by 0.5 days and raise recovery score to ≥ 75.0 within 2 quarters.
2	Cardiology Protocol Review	Commission a clinical audit benchmarked against Dermatology and General Medicine practices. Target: raise recovery score to ≥ 75.0 within 2 quarters.
3	Physician Peer-Learning Programme	Pair Dr. Johnson and Dr. Miller as mentors for physicians below 74.0 recovery. Separately investigate Dr. Martinez's fast-discharge model (4.21 days) for replicable best practices.
4	Surgical Bed Occupancy Management	Surgery patients average 10.07 days — 4.8x longer than Observation. Review surgical referral criteria for cases where Therapy could achieve comparable outcomes at significantly lower bed-day cost.
5	Observation Discharge Protocol	2.09-day average stay + 72.86 recovery score signals premature discharge risk. Introduce readmission-risk scoring before all Observation discharges.
6	Pediatrics Cost-Outcome Review	Highest average cost (£5,588) with only average recovery (75.46). A cost-reduction task force should identify inefficiencies without compromising clinical outcomes.
7	Power BI Real-Time Monitoring	This dashboard and analysis framework can be extended into Power BI for real-time hospital performance monitoring and predictive analytics, enabling proactive management of outlier recovery cases and early cost-overflow alerts.
8	LinkedIn Portfolio Write-Up	

The following write-up is ready for direct publication on LinkedIn as a portfolio piece.

From Raw Data to Clinical Insight: Hospital Recovery Analysis Dashboard | Excel + Data Analytics

I recently completed a full end-to-end data analytics project for Green Valley Medical Center — transforming 400 raw patient records into a fully interactive Excel dashboard that surfaces actionable clinical and operational insights for stakeholders.

What I built:

A multi-sheet Excel workbook featuring an interactive slicer-driven dashboard, pivot table analyses, calculated KPI metrics, and a professionally formatted stakeholder report — all from a single raw dataset.

Key insights uncovered:

- Dermatology leads all departments in recovery (76.61) with the shortest average stay (4.67 days) — proving clinical efficiency and quality go hand in hand.
- Surgery patients average 10.07 days in hospital — nearly 5x longer than Observation (2.09 days) — the largest single bed-occupancy challenge in the system.
- Dr. Johnson tops recovery performance (78.51) while Dr. Martinez achieves the fastest patient throughput (4.21 days) — two different and equally valuable dimensions of clinical excellence.
- 80% of patients fell into the Low Cost category yet produced outcomes comparable to the High Cost group.
- Teenagers have both the lowest recovery scores (73.93) AND the longest hospital stays (5.90 days) — the highest-priority demographic for pathway redesign.
- Elderly patients (76.03 recovery, 5.41 avg stay) outperform multiple younger age groups on both metrics.

Tools used: Microsoft Excel (Pivot Tables, Slicers, Dynamic Charts, IFS, Named Ranges), Data Cleaning, Dashboard Design, Stakeholder Report Writing.

The broader vision:

This dashboard and analysis framework can be extended into Power BI for real-time hospital performance monitoring and predictive analytics — enabling leadership to proactively track outlier recovery cases, forecast cost pressures, and drive evidence-based clinical decisions before issues escalate.

This project deepened my ability to bridge the gap between raw data and strategic decision-making — turning numbers into narratives that hospital leadership can act on immediately.

#DataAnalytics #HealthcareAnalytics #Excel #DashboardDesign #PowerBI #HospitalData #RecoveryAnalysis
#DataStorytelling #GreenValleyMedicalCenter #PortfolioProject

9 Conclusion

The verified Green Valley Medical Center dataset reveals a clear hierarchy of performance across departments, treatment types, physicians, and demographics. The headline finding remains that Dermatology and General Medicine lead recovery outcomes while Cardiology and Neurology lag — but this analysis now adds a critical dimension: **Dermatology achieves top recovery with the shortest patient stays**, providing a model of clinical efficiency that other departments should benchmark against.

The hospital stay data further reveals that **Teenagers — not Elderly patients — are the most challenging demographic**, combining the worst recovery scores with the longest stays. Elderly patients demonstrate that high-quality care intensity can deliver above-average outcomes efficiently (76.03 recovery, 5.41 days).

Physician throughput analysis confirms that fast discharge (Dr. Martinez, 4.21 days) and high recovery (Dr. Johnson, 78.51) represent distinct clinical strengths. Both should be case-mix adjusted, validated, and shared across the physician cohort through a structured peer-learning programme.

Extending this analytical framework into Power BI for real-time monitoring would enable Green Valley Medical Center to move from retrospective reporting to proactive, predictive hospital performance management — a critical capability for sustainable quality improvement at scale.

Report prepared by **Funmilola Lanade** | Green Valley Medical Center Data Analytics | April 2026